CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (ME,CL)

November – December 2011

CL 202 Strength of Materials

Date: 30.11.2011, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 01.** **[07]**

Q - 2 (a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 02.** **[07]**

(b) A rectangular wooden beam of size 200 mm × 300 mm is strengthened by steel plates of 10 mm thickness covering entire width of wooden section at top and bottom. Find the moment carrying capacity of the composite section if allowable stresses in wood and steel are 20 MPa and 100 MPa respectively. Take modular ratio as 10. **[04]**

(c) A simply supported beam of 6 m span is subjected to uniformly distributed load of 15 kN/m throughout the span. The cross-section of rectangular beam is 300 mm × 600 mm. Find the maximum bending stress in the cross-section of beam. Calculate the bending stress at a distance of 2 m from the left hand support. **[03]**

OR

Q - 2 (a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 03.** **[07]**

(b) For a cantilever beam loading and cross-section are as shown in **Figure 04**. Calculate **[04]**

(a) The bending stress at the point of maximum bending moment.

(b) Radius of curvature of neutral layer where bending moment is maximum.

(c) Bending stress at 50 mm below top surface of cross-section.

(c) A beam is having cross-section shown in **Figure 05.** If the maximum stress in the beam section is not to exceed 100 N/mm^2. Determine the maximum bending moment the beam can carry safely. **[03]**

Q - 3 (a) Prove with usual notations: $\frac{f}{y} = \frac{E}{R}$ **[03]**

(b) State and prove the perpendicular axis theorem. **[04]**

(c) Find moment of inertia of lamina shown in **Figure 06** about XX axis. **[07]**

OR

Q - 3 (a) A water main of 1000 mm internal diameter and 10 mm thickness is running full as shown in **Figure 07**. If bending stress is not to exceed 70 N/mm^2, find the greatest span on which the pipe may be freely supported. Density of steel and water is 78000 N/m^3 and 10000 N/m^3 respectively. **[03]**

(b) Determine moment of inertia of area shown in **Figure 08** about its base B-B. **[04]**

(c) Find moment of inertia of lamina shown in **Figure 09** about its horizontal and vertical centroidal axis.

SECTION – II

Q - 4 (a) Write the general equation of torsion. Define polar moment of inertia and find the same for a solid circular section having 200mm diameter. **[04]**

(b) Draw qualitative shear stress distribution diagrams (shape only) for the following cross-sections. **[03]**

i. T section
ii. Hollow rectangle
iii. L section

Q - 5 (a) Define principal plane and state the importance of finding the same. Draw Mohr's stress circle for the element subjected to the stresses as shown in **Figure 10** and find: (i) Major and minor principal stresses (ii) location of principal planes and (iii) maximum shear stress and its location. **[07]**

(b) For an element shown in **Figure 11** and find: (i) Major and minor principal stresses (ii) location of principal planes and (iii) maximum shear stress and its location. **[07]**

OR

Q - 5 (a) Find (i) Normal and shear stress at a plane located at 60° with vertical, and (ii) maximum shear stress and its location, for the element shown in **Figure 12.** **[07]**

(b) Find (i) Major and minor principal stresses (ii) location of principal planes for the element shown in **Figure 12.** **[07]**

Q - 6 (a) A solid circular shaft is required to transmit 100 kW at 100 rpm. Find the diameter of the shaft (i) If the maximum shear stress in the material is 70 MPa and (ii) allowable angle of twist per meter length is 4°. Assume G = 60 GPa. **[07]**

(b) A beam is having square cross-section with one diagonal horizontal as shown in **Figure 13.** If it is subjected to maximum shear force of 100 kN, draw shear stress distribution diagram at N.A. **[07]**

OR

Q - 6 (a) Find the power transmitted by a hollow circular shaft of external diameter of 150mm and internal diameter of 100mm running at 200 rpm. The maximum shear stress induced in the shaft is 70 MPa. **[07]**

(b) Derive q = SAY / BI with usual notations. **[07]**

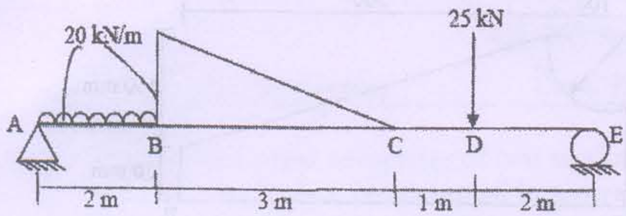

Figure 1

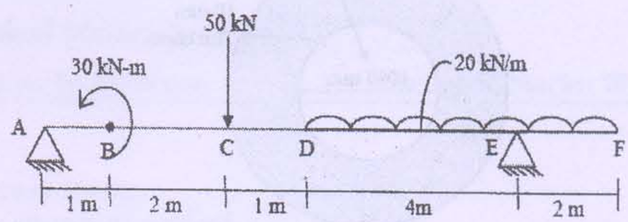

Figure 2

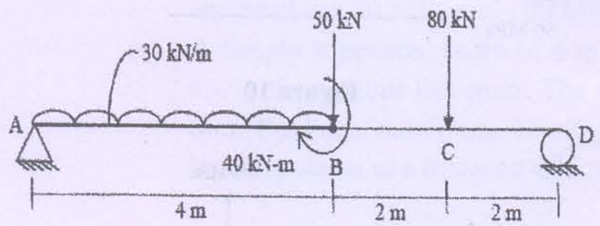

Figure 3

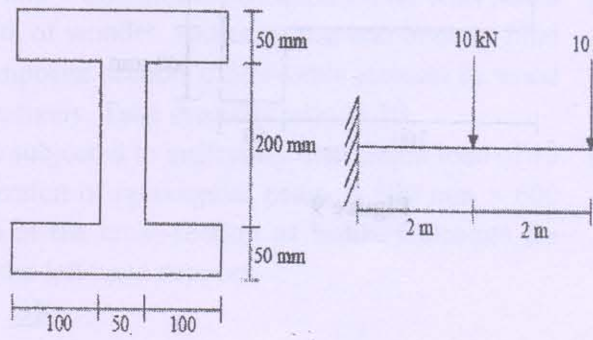

Figure 4

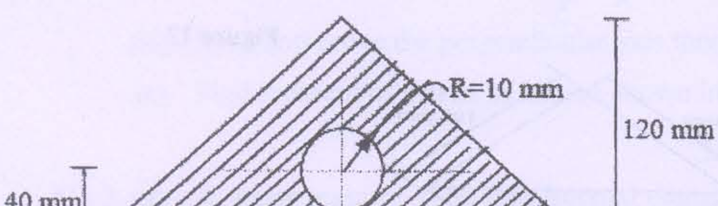

Figure 5

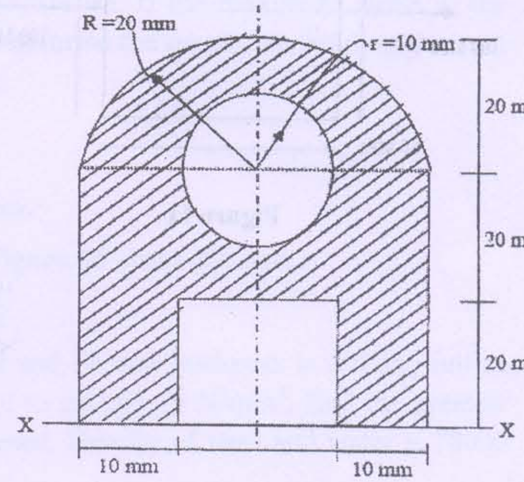

Figure 6

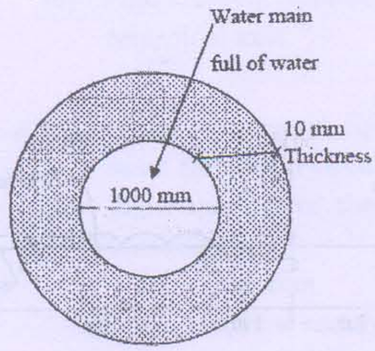

Figure 7

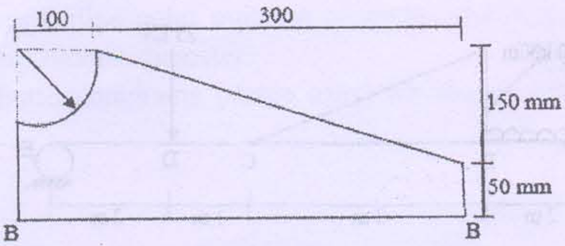

Figure 8

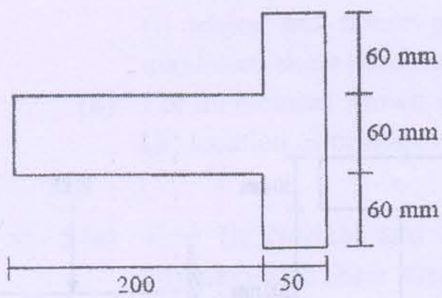

Figure 9

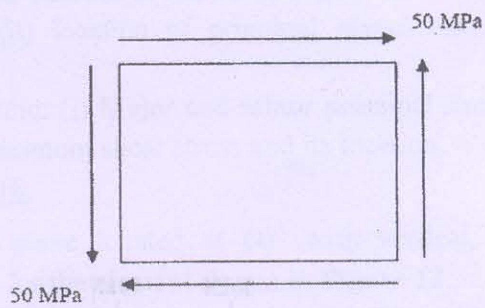

Figure 10

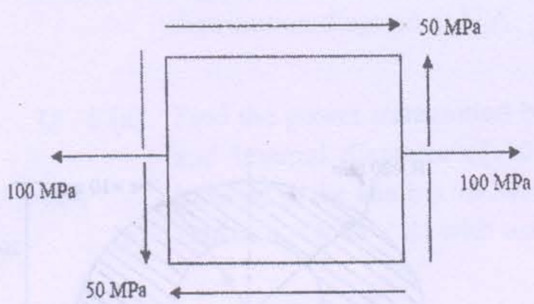

Figure 11

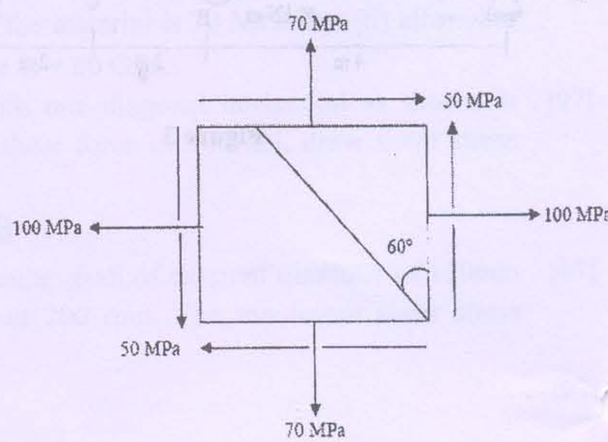

Figure 12

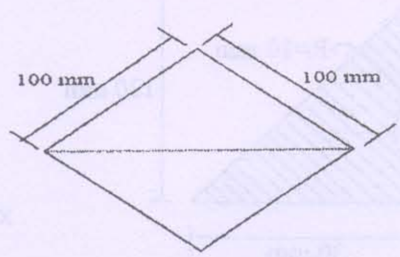

Figure 13

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